



Loan Prediction Project

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Team 4

Project Understanding

What Problem Are We Solving + End Goal

Problem

- Predicting loan approval to help financial institutions manage risks.
- Addressing imbalances in approval rates by identifying key factors that distinguish approved and rejected applications.

End Goal

- Build a machine learning model that predicts loan approval using features like applicant's income, employment length, credit history, etc.
- Provide financial institutions with data-driven insights to enhance the *fairness* and *efficiency* of the loan approval process while minimizing defaults.

Dataset Overview

Observations

Simulated dataset
containing 32,581 loan
applications

- Predictor Variables
 - 7 numeric
 - 4 categorical
- Target variable:
 - Rejected: 0
 - Approved : 1

Numeric Variables

- Age
- Income
- Employment length
- Credit history length
- Loan Amount
- Loan Interest Rate
- Percentage of Money
allocated towards loan
repayment

Categorical Variables

- Home ownership type
- Purpose of loan
- Loan Grade type
- Default history

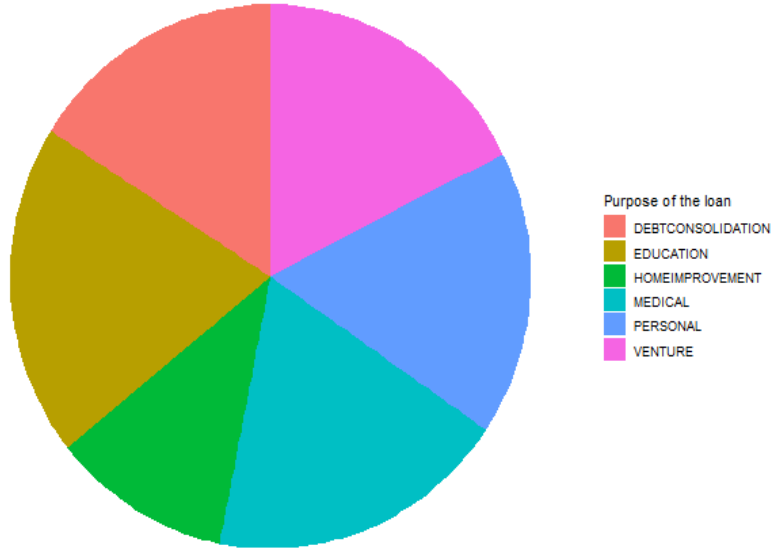
Variable Summaries

variable	n	missing	missing_pct	unique	unique_pct	mean	min	Q1	median	Q3	max	sd
person_age	32581	0	0.00	58	0.178	27.735	20.00	23.00	26.00	30.00	144.00	6.348
person_income	32581	0	0.00	4295	13.183	66074.848	4000.00	38500.00	55000.00	79200.00	6000000.00	61983.119
person_emp_length	32581	895	2.75	37	0.114	4.790	0.00	2.00	4.00	7.00	123.00	4.143
loan_amnt	32581	0	0.00	753	2.311	9589.371	500.00	5000.00	8000.00	12200.00	35000.00	6322.087
loan_int_rate	32581	3116	9.56	349	1.071	11.012	5.42	7.90	10.99	13.47	23.22	3.240
loan_status	32581	0	0.00	2	0.006	0.218	0.00	0.00	0.00	0.00	1.00	0.413
loan_percent_income	32581	0	0.00	77	0.236	0.170	0.00	0.09	0.15	0.23	0.83	0.107
cb_person_cred_hist_length	32581	0	0.00	29	0.089	5.804	2.00	3.00	4.00	8.00	30.00	4.055

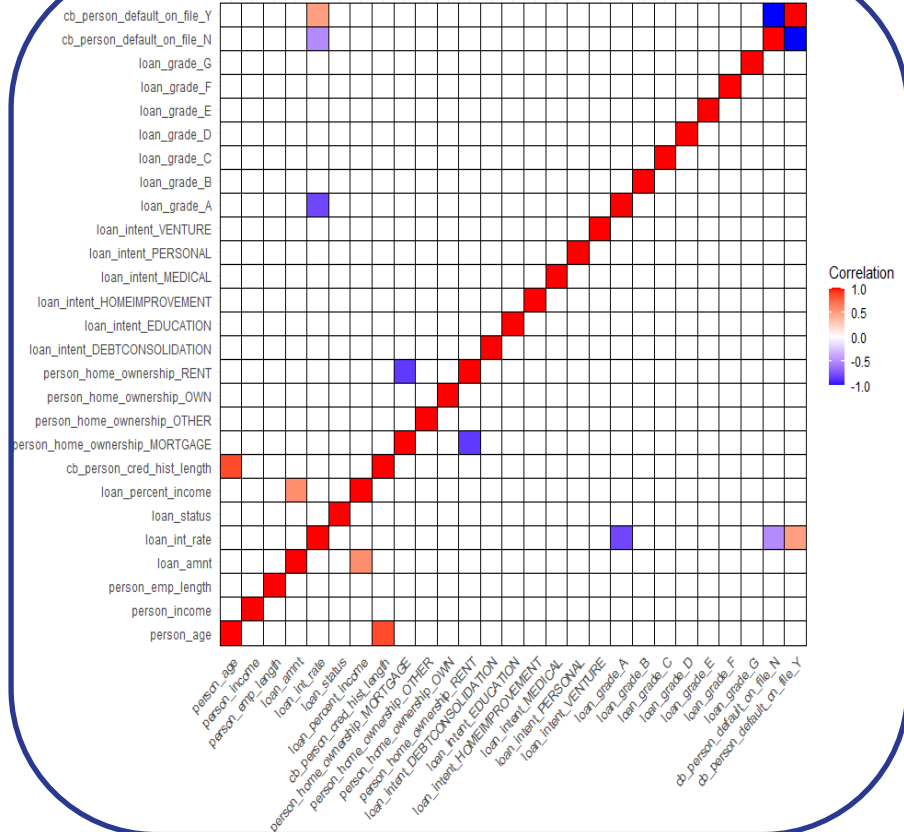
variable	n	missing	missing_pct	unique	unique_pct	freqRatio	1st mode	1st mode freq	2nd mode	2nd mode freq	least common	least common freq
person_home_ownership	32581	0	0	4	0.012	1.22	RENT	16446	MORTGAGE	13444	OTHER	107
loan_intent	32581	0	0	6	0.018	1.06	EDUCATION	6453	MEDICAL	6071	HOMEIMPROVEMENT	3605
loan_grade	32581	0	0	7	0.021	1.03	A	10777	B	10451	G	64
cb_person_default_on_file	32581	0	0	2	0.006	4.67	N	26836	Y	5745	Y	5745

Data Understanding

Distribution of Loan Intent



Heatmap of Filtered Correlation Matrix ($|\text{correlation}| > 0.5$)





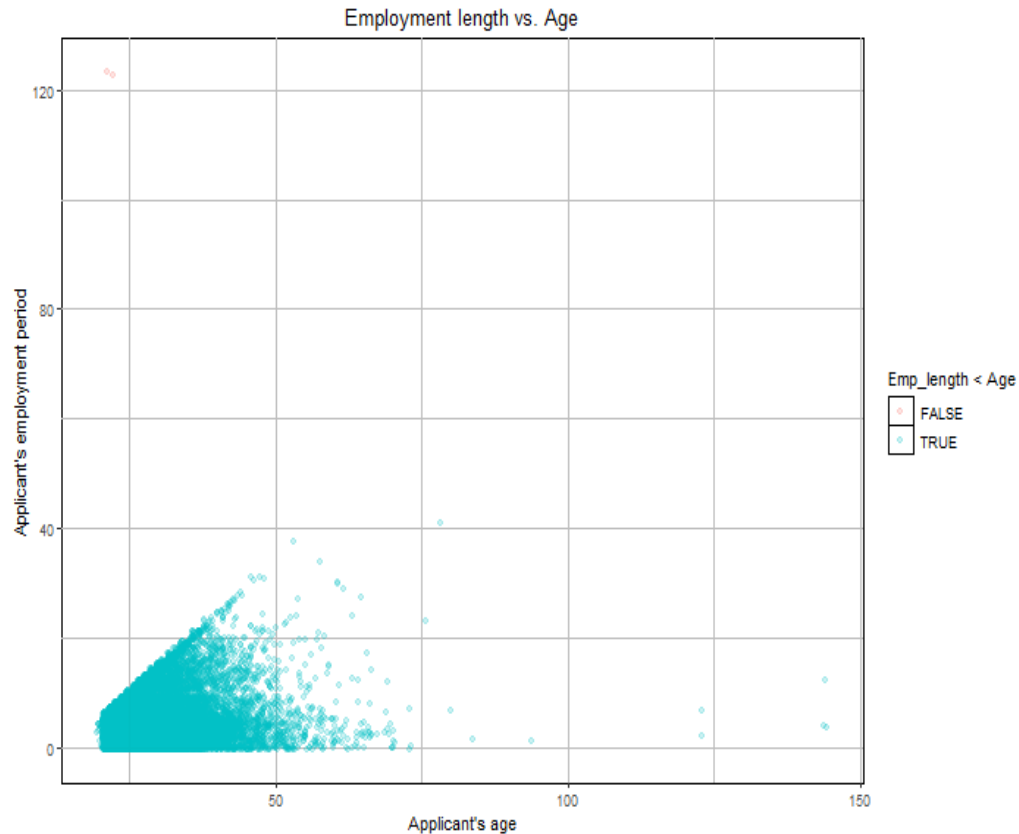
Data Preparation for supervised learning

Missing Value Imputation

- Only two variables `applicant's employment length` (2.75%) and `loan interest rate` (9.56%) have missing values.
- The missing values are imputed using predictive mean matching (PMM). technique
 - Relevant variables for `person_emp_length` : `person_age`, `person_income`, and `cb_person_cred_hist_length`
 - Relevant variable for `loan_int_rate` : `loan_amnt`, `person_income`, `loan_grade`, and `cb_person_cred_hist_length` .

Handling Outliers

- Outliers can skew results and impact model performance.
- Outliers removed:
 1. Borrower's age > 120 years
 2. Employment length > borrower's age



Feature Transformation

Log transformation

- Applied to skewed numerical variables
- Variables affected:
 - ❑ `person_age`
 - ❑ `person_emp_length`
 - ❑ `loan_amnt`
 - ❑ `loan_int_rate`

Feature engineering

Helps enhance a model's ability and improves predictive performance.

New feature was introduced:

- *loan_percent_income* as ratio of loan amount and applicant's income

Standardization

- Standardized all the numerical variables such that mean = 0 and standard deviation = 1

Modelling

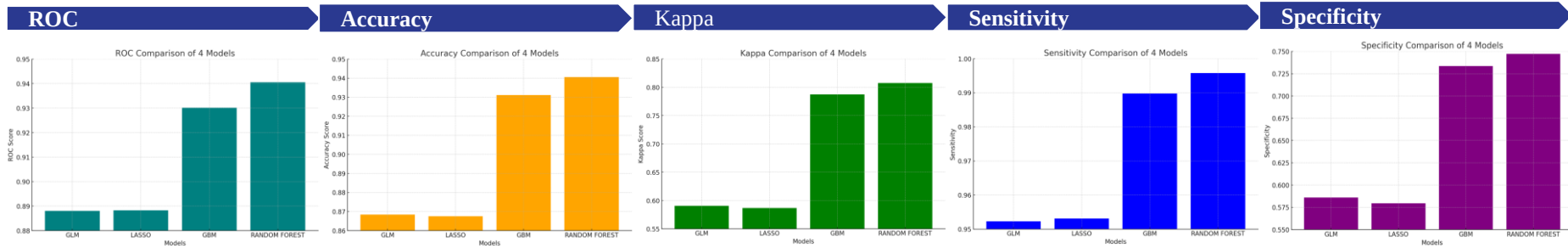
Supervised Learning



OBJECTIVE : Predict loan application

CANDIDATE MODELS : GLM, LASSO, GBM, RANDOM FOREST, all evaluated using cross-validation

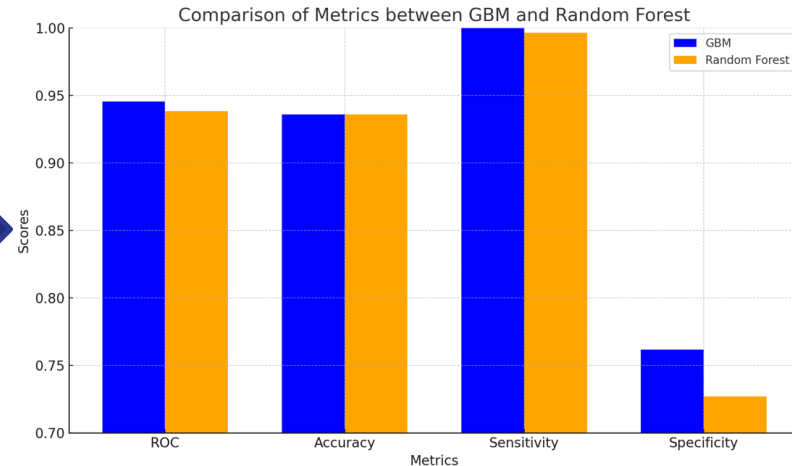
FIRST-ROUND modeling :



SECOND-ROUND modeling :

■ **TOP MODEL:** GBM + 10-fold cross-validation (CV) + random search hyperparameter tuning

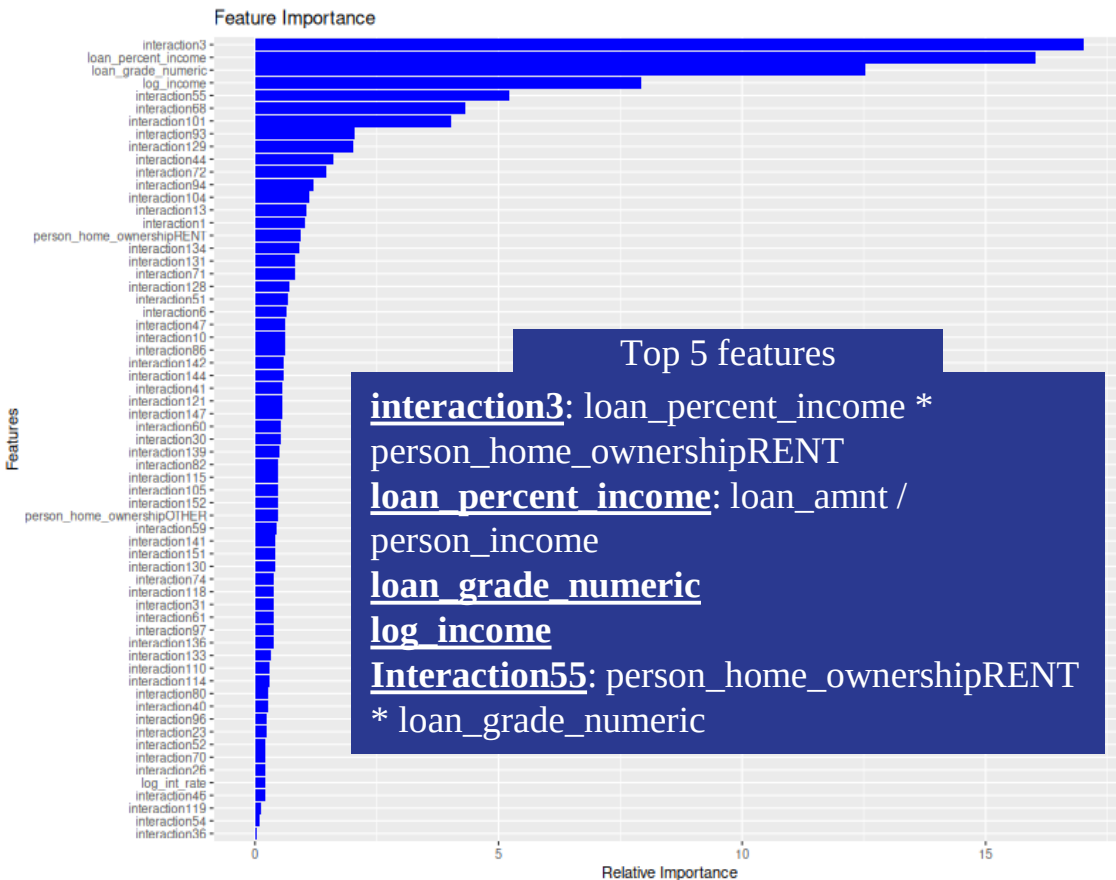
1. Both GBM and Random Forest demonstrate high modeling quality
2. GBM slightly outperforms in all metrics, particularly excelling in positive class recognition (Sensitivity) and negative class recognition (Specificity)
3. Both are highly reliable in overall classification performance
4. Both models achieve an accuracy of over 93%, indicating a low error rate in overall predictions and making the results highly reliable
- 5 Although the specificity of both models is relatively good, there is still room for improvement (GBM: 0.76, Random Forest: 0.73)



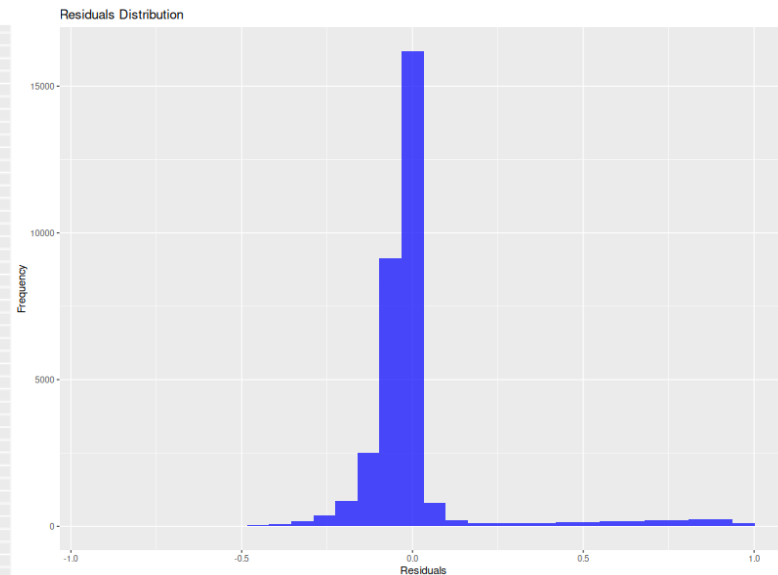
Analysis Results



Feature Importance



Residuals Distribution



Overview and Improvement of the Optimal Model

METRICS	VALUE	COMMENT
Accuracy	0.9531	The high accuracy indicates that the model performs well overall
Kappa	0.852	The model demonstrates high reliability in predicting both classes ("No" and "Yes")
Sensitivity	0.9976	The model's ability to identify the positive class ("No") is very strong, with almost no missed predictions.
Specificity	0.7936	The model's ability to identify the negative class ("Yes") is relatively weak, and the false positive rate still needs improvement
PPV	0.9454	The model's predictions for the positive class are highly reliable, with only about a 5.46% false positive rate
NPV	0.9895	The model is highly reliable in predicting negative samples, with only about a 1.05% false rejection rate
Balanced Accuracy	0.8956	The model's classification ability for both positive and negative samples is well-balanced, and its overall performance is good

IMPROVEMENT

1. Class imbalance issue - **Introduce class weights into the model to impose a higher penalty on the "Yes" classification**
2. False Negatives - **Adjust the classification threshold to reduce false negatives, Use more complex models (such as XGBoost or deep learning) to capture the patterns of "Yes"**
3. McNemar's Test P-Value: The P-value is significantly less than 0.05, indicating a significant asymmetry between the model's classification errors (False Positives and False Negatives) **Particular attention should be paid to balancing false positives and false negatives**

MODEL ADVANTAGES

- High accuracy (95.31%) and a high Kappa value (0.852) indicate that the model performs well overall
- The model performs exceptionally well in predicting “No,” with almost no missed cases (Sensitivity = 0.9976)

MODEL DISADVANTAGES

- The prediction ability for "Yes" (Specificity = 0.7936) is relatively insufficient
- Class imbalance has impacted the model’s ability to effectively distinguish “Yes”
- IMPROVEMENT: Balance the class distribution, adjust the classification threshold, and focus on optimizing the "Yes" category

Unsupervised Learning

DBSCAN Clustering

DBSCAN Clustering Advantages

- No need to specify the number of clusters
 - Works well for imbalanced clusters
 - Robust to noise and outliers
 - No strict assumptions about data distribution
-

Data PreProcessing

Missing Values

Imputation

- Missing values corresponding to two variables: person's employment length and loan interest rate
- 5-NN imputation used.

Outliers

Removal

- Two data points where person's employment length is greater than person's age. This is practically impossible.

Feature selection

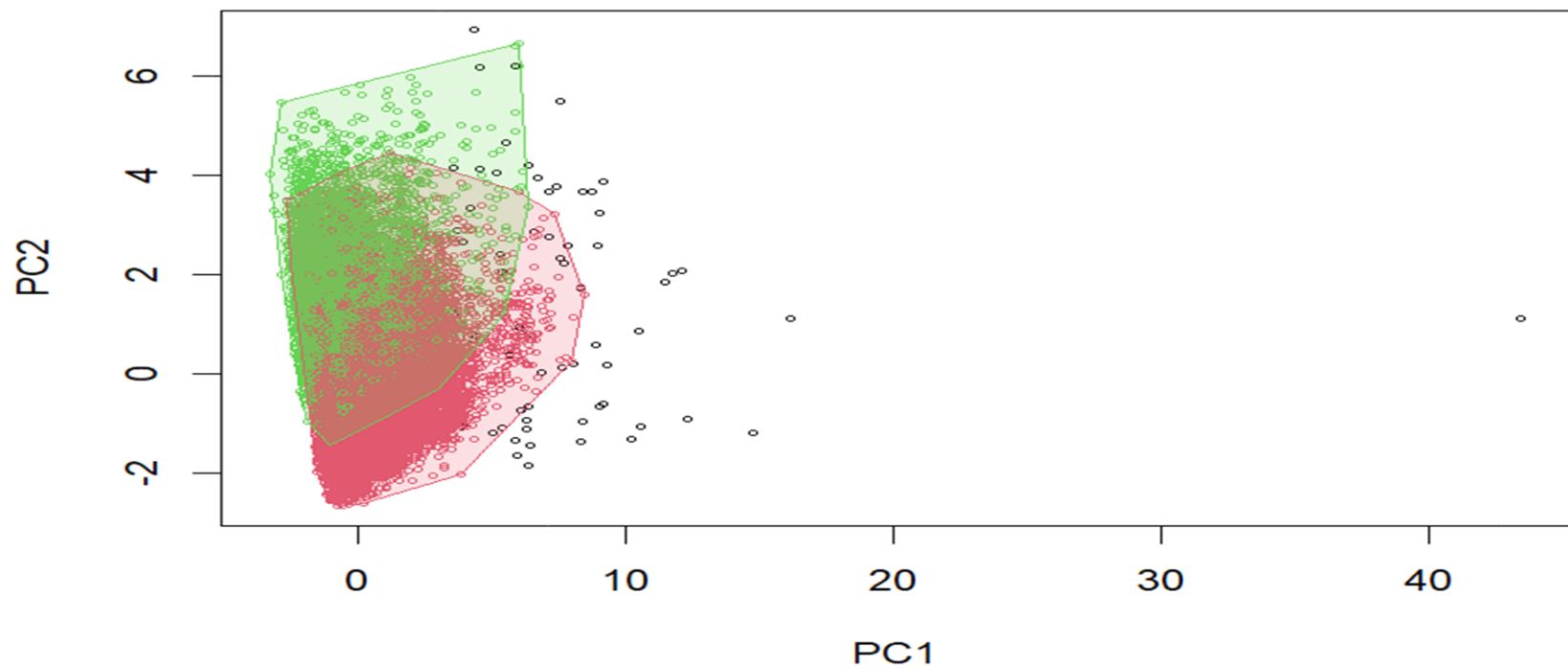
Selection

- Only numeric features are selected

Transformation

- Variables standardized to have a mean of 0 and standard deviation of 1.

Convex Cluster Hulls



Summary

Project Understanding

Data Understanding

Data Preparation

Modelling

Evaluation

The problem is to evaluate loan approval process and predict which applications will be approved or rejected.

Dataset containing 7 numeric, 4 categorical predictors and 1 target variable.

Missing values for applicant's employment length and loan interest rate. Two outlier points where applicant's age is less than their employment period are removed.

For Supervised learning: PMM for missing value imputation.
For Unsupervised learning: kNN (k=5) for missing value imputation.

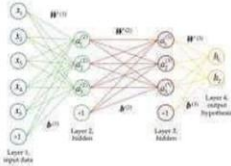
Supervised learning: Gradient Boosting Machine (GBM) has better predictability among other models considered.

Unsupervised learning: DBSCAN clustering used to find two clusters and some noise points that need further investigation.

GBM Model: Specificity = 0.7936, Sensitivity = 0.9976

DBSCAN clustering: ARI = 0.994, F1 Score =1, Precision=1, Recall=1

Thank you for listening fellow parrots!

	A parrot	Machine learning algorithm
		
Learns random phrases	✓	✓
Doesn't understand about what it learns	✓	✓
Occasionally speaks nonsense	✓	✓
Is a cute birdie parrot	✓	✗